

## MATS 1112/1182 Study Guide

### Subsolidus Equilibrium Compatibility Triangles

The phases present at equilibrium are defined by the phases located at the corners of the subsolidus equilibrium compatibility triangle. That is, for any composition located within such a triangle, the phases on the corners are the equilibrium phases present.

### Invariant Points

Every subsolidus equilibrium compatibility triangle has its own invariant point and this invariant point defines the temperature (and composition) at which the liquid first forms for this mixture of solid phases. The invariant point is defined by the intersection of the three primary crystallisation fields that correspond to the phases on the corners of the subsolidus equilibrium compatibility triangle.

### Melting Points

A *congruently melting* phase is one that undergoes a phase transformation from a solid to a liquid without changing its composition; this is a true melting point. An *incongruently melting* phase is one that undergoes a phase transformation from a solid of one composition to a solid and liquid, both of composition different from that of the original; this is a peritectic decomposition.

The composition of a *congruently melting* phase lies within its own primary crystallisation field. The convention is that congruent melting points generally are given under the composition. The composition of an *incongruently melting* phase lies outside of its own primary crystallisation field. The convention is that incongruent melting points are not indicated on the phase diagram.

In the absence of its being specified on the phase diagram, the peritectic decomposition temperature of a phase can be determined by noting the invariant point of the highest temperature on the boundary lines that encircle its primary crystallisation field. When this temperature is exceeded, then the liquid region encompasses the primary crystallisation field and it disappears. In its absence, solid-liquid equilibrium no longer can exist. Therefore, the solid cannot exist either.

### Eutectics and Peritectics

Eutectics lie within their own subsolidus equilibrium compatibility triangles. Hence, the liquid can remain in equilibrium with the three solid phases on the corners of the triangle without changing the pre-existing subsolids phase relations.

Peritectics lie outside of their own subsolidus equilibrium compatibility triangles. When a peritectic transformation occurs, it causes one subsolidus join to disappear and another to appear in order to allow all three solid phases to remain in equilibrium with the liquid.

### Lever Rule

The *binary lever rule* requires a line joining two phases between which the composition of interest lies. The relative proportions of each of the two phases are determined by the relative (inverse) lengths of the segments of the line divided by the composition of interest.

The *ternary lever rule* requires three lines within a triangle, each line being drawn from a corner of the triangle through the composition of interest to the opposite side of the triangle. The relative proportions of each of the three phases are determined by the relative (inverse) lengths of the segments of the lines divided by the composition of interest.

With an equilateral triangle, the proportions of the phases can be read directly from the (100%) divisions marked on the diagram by drawing lines parallel to the triangle edges through the composition of interest and reading the (inverse) percentages.

## Isothermal Sections

Isothermal sections consist of a) single-phase solid and liquid solution homogeneity regions (0 dimension), b) wedges between two phases, typically solid-liquid or solid solution-solid solution (1 dimension), and c) triangles of three phases (2 dimensions).

Isothermal sections must be completed in three stages: a) identify the single-phase liquid solution homogeneity region(s), b) complete the solid-liquid equilibria, and c) repeat the pre-existing solid-solid equilibria.

For a), outline the isotherms relevant to the temperature of interest. The area within this region is the liquid solution homogeneity region.

For b), the primary crystallisation field across which the liquidus surface crosses must be identified. Since this defines the solid that is in equilibrium with the liquid, then the wedge must be joined from this solid to the liquid in this region.

For c), the pre-existing subsolidus compatibility tie-lines must be repeated. Be careful when the liquid region envelopes the entire primary crystallisation field of a phase. When this happens, the solid no longer can exist and so there can be no tie-lines to this phase. In particular, there will be at least one solid-solid tie-line that is crossed by the liquid region, so it is implicit that this pre-existing tie-line must disappear. The remaining two solid-solid tie-lines joined to this phase then also must disappear because their continued existence would result in a non-triangular (or non-wedged) arrangement of tie-lines.

## Crystallisation Path

When a composition is completely melted and then cooled so as to crystallise it, it is possible to follow the change in composition of the liquid during cooling. The first solid to crystallise or precipitate is that defined by the primary crystallisation field in which the composition of interest lies. The composition of the liquid, of course, starts at that specific composition. If, for example, this composition is in the *A* primary crystallisation field, then *A* precipitates and is removed from the solution (single-phase precipitation). Therefore, the composition of the liquid must be depleted in *A* and so the liquid composition moves away from the composition of *A*.

Upon further cooling, the liquid composition moves downhill along the liquidus surface until it reaches a relative minimum at the boundary line. The boundary line joins two primary crystallisation fields and so the two solids defined by these fields must be in equilibrium with the liquid. Therefore, if, for example, the boundary line joins the primary crystallisation fields of *A* and *B*, then these two solids must precipitate simultaneously as the liquid composition moves downhill along the boundary line (two-phase precipitation).

Upon further cooling, the liquid composition moves downhill along the boundary line until it reaches a relative minimum at an invariant point. If the invariant point is that which belongs to its parent composition, then the process stops. That is, if the invariant point, defined by the intersection of three primary crystallisation fields, matches the three corners of the subsolidus equilibrium compatibility triangle, then the process stops.

If the invariant point does not match the subsolidus equilibrium compatibility triangle, then the process continues until the matching invariant point is reached. In this case, the boundary lines traversed by the liquid must be *resorption* curves, during which one of the precipitated phases redissolve. Such phases cannot be present at final equilibrium since they are not present in the original composition's parent subsolidus equilibrium compatibility triangle.

When the correct invariant point is reached, three phases precipitate simultaneously. If, for example, the parent subsolidus equilibrium compatibility triangle consisted of *A*, *B*, and *C*, then these three phases precipitate simultaneously at the invariant point temperature and the process ceases (three-phase precipitation).

## **MATS2008/3002 Phase Equilibria – Processing Definitions**

### **Solid-State Sintering or Densification**

This is a high-temperature sintering method that relies on solid-state diffusion to cause mass transfer that results in densification. There is no liquid involved.

### **Liquid-Phase Sintering or Densification**

For very stable materials for which the diffusion coefficients are low, the alternative high-temperature sintering method relies on the presence of a small amount of liquid to enhance particle rearrangement, solution-precipitation, and interparticle bonding. In ceramics, this liquid often is a glass, which does not recrystallise upon cooling and so remains on the grain boundaries. In some cases, the liquid may be an invariant point liquid, which recrystallises upon cooling. In the latter case, this is known as *transient liquid-phase sintering*. Both types of residual grain-boundary phases will affect the properties, especially at high temperatures.

### **Casting**

This is shape forming achieved by the complete melting of a material and pouring into a mould. The melt should be homogeneous and so it must be congruently melting.

### **Czochralski Single-Crystal Growth**

This involves positioning a single-crystal above a homogeneous melt of the same composition, allowing the melt to extend (initially by vapour-phase diffusion) onto the single crystal, and rotating and withdrawing the single crystal away from the melt. A larger, continuous, and oriented single crystal then can be pulled out of the melt, where the temperature drop from the thermal gradient is sufficient to prevent melting of the large single crystal above the melt. The melt should be homogeneous and so it must be congruently melting.

### **Flux-Growth of Single Crystals**

In cases in which the material is not congruently melting, then flux-growth is the normal alternative to produce small, individual, single crystals. This involves heating (1) an incongruently melting composition or (2) a mixture that contains an incongruently melting composition either above (*i.e.*, causing precipitation) or below (*i.e.*, causing grain growth) the liquidus temperature. The single crystal of interest is the primary phase, so it is compatible with the liquid (*viz.*, the flux) and so does not react with the single crystal. Therefore, after solidification, the mixture will consist of single crystals embedded in a solidified matrix of recrystallised liquid. Ideally, a solvent can be used to dissolve the matrix but not the single crystals. If there is no suitable solvent, then the single crystals must be separated from the matrix by hand or other mechanical means. The slower the cooling rate, the larger the single crystals. Note that flux-growth also can be done with congruently melting compounds because the requisite solid + liquid phase field is present.

## Classification of Binary Phase Equilibrium Diagrams

### Simple Eutectic

#### Polymorphism

*Subsolidus*

*Supersolidus*

#### With Intermediate Compound

*Congruent Melting*

*Incongruent Melting*

Peritectic

Inverse Peritectic

*Dissociation*

Upper Temperature Limit of Stability (Peritectoid)

Lower Temperature Limit of Stability (Eutectoid)

Upper and Lower Temperature Limits of Stability

*Polymorphism*

Subsolidus

Supersolidus

*Degenerate Eutectic*

### Solid Solution

#### Primary

*Complete Miscibility*

Continuous

Ascending

Descending

Maximal Melting Point

Minimal Melting Point

With Miscibility Gap

With Superlattice

*Partial Miscibility*

Eutectic

Peritectic

Eutectoid

Peritectoid

With Intermediate Compound

Congruent Melting

Incongruent Melting

Peritectic

Inverse Peritectic

Eutectoid

Peritectoid

#### Secondary

### Liquid Immiscibility

#### Increasing Mutual Solubility

*Monotectic*

*Syntectic*

*Metastable Immiscibility*

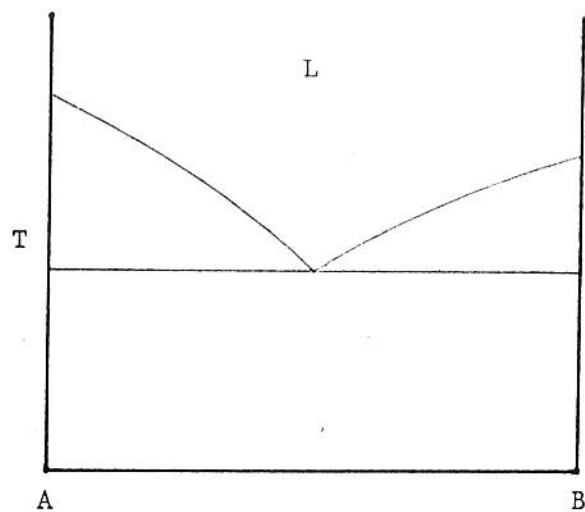
*Spinodal Decomposition*

#### Decreasing Mutual Solubility

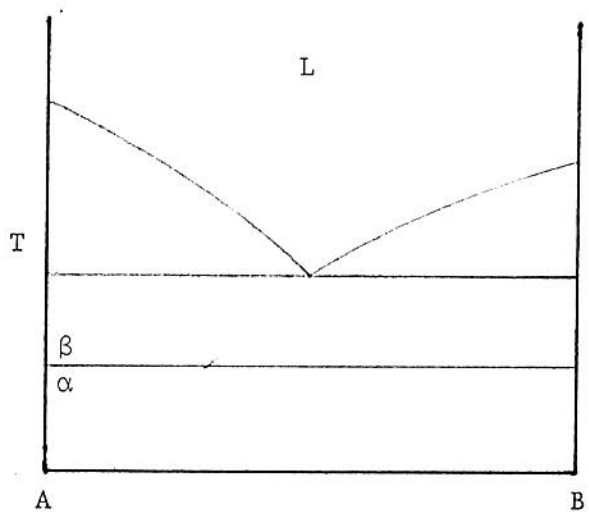
**Closed Solubility**

## Classification of Binary Invariant Points (T-X Diagrams)

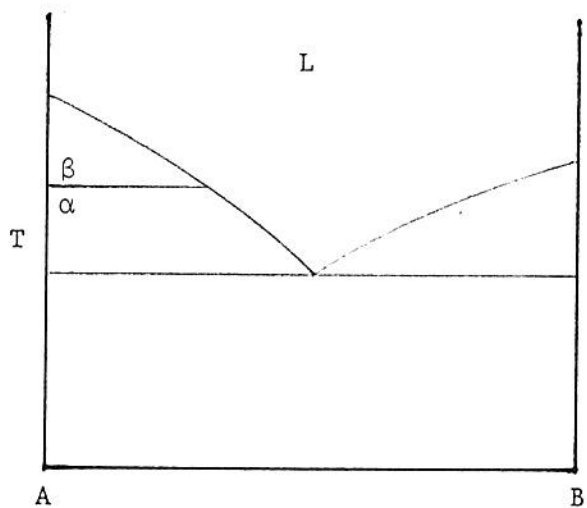
| Eutectic-Type ( $P_1 \rightarrow P_2 + P_3$ )                                                                                                                                                                                                                                                                                          |                             | Upon Cooling                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------------------|
| Eutectic                                                                                                                                                                                                                                                                                                                               |                             | $L \rightarrow S_1 + S_2$        |
| Eutectoid                                                                                                                                                                                                                                                                                                                              | (Downward Dissociation)     | $S_1 \rightarrow S_2 + S_3$      |
| Monotectic                                                                                                                                                                                                                                                                                                                             | (Consolute Point – Liquid)  | $L_1 \rightarrow L_2 + L_3$      |
| Monotectoid                                                                                                                                                                                                                                                                                                                            | (Consolute Point – Solid)   | $S_1' \rightarrow S_2' + S_3'$   |
| Inverse Peritectic                                                                                                                                                                                                                                                                                                                     | (Catatectic)                | $S_1 \rightarrow S_2 + L_3$      |
| Spinodal Decomposition                                                                                                                                                                                                                                                                                                                 | (Critical Point – Liquid)   | $L_1' \rightarrow L_2' + L_3'$   |
| Spinodal Decomposition                                                                                                                                                                                                                                                                                                                 | (Critical Point – Solid)    | $S_1 \rightarrow S_2 + S_3$      |
| Peritectic-Type ( $P_1 \rightarrow P_2 + P_3$ )                                                                                                                                                                                                                                                                                        |                             | Upon Heating                     |
| Peritectic                                                                                                                                                                                                                                                                                                                             | (Incongruent Melting Point) | $S_1 \rightarrow S_2 + L_3$      |
| Peritectoid                                                                                                                                                                                                                                                                                                                            | (Upward Dissociation)       | $S_1 \rightarrow S_2 + S_3$      |
| Syntectic                                                                                                                                                                                                                                                                                                                              |                             | $S_1 \rightarrow L_2 + L_3$      |
| Crystallographic Phase Transformations                                                                                                                                                                                                                                                                                                 |                             | Upon Heating or Cooling          |
| Subsolidus                                                                                                                                                                                                                                                                                                                             |                             | $\alpha_1 \rightarrow \beta_1$   |
| Supersolidus                                                                                                                                                                                                                                                                                                                           | (Proeutectic)               | $\alpha_1 \rightarrow \beta_1$   |
| Superlattice                                                                                                                                                                                                                                                                                                                           |                             | $\alpha_1 \rightarrow \alpha_1'$ |
| State Phase Transformations                                                                                                                                                                                                                                                                                                            |                             | Upon Heating or Cooling          |
| Melting Point                                                                                                                                                                                                                                                                                                                          | (Congruent Melting Point)   | $S_1 \rightarrow L_1$            |
| Degenerate Eutectic                                                                                                                                                                                                                                                                                                                    |                             | $S_1 \rightarrow L_1$            |
| Boiling Point                                                                                                                                                                                                                                                                                                                          |                             | $L_1 \rightarrow G_1$            |
| Sublimation Point                                                                                                                                                                                                                                                                                                                      |                             | $S_1 \rightarrow G_1$            |
| Triple Point (T-P Diagrams)                                                                                                                                                                                                                                                                                                            |                             | Unique Condition                 |
| Solid + Liquid + Gas Phases Coexist Simultaneously                                                                                                                                                                                                                                                                                     |                             |                                  |
| Key                                                                                                                                                                                                                                                                                                                                    |                             |                                  |
| T = Temperature<br>X = Composition<br>P = Phase<br>G = Gas<br>L = Liquid<br>S = Stoichiometric Solid or Solid Solution<br>S = Stoichiometric Solid<br>S' = Solid Solution<br>$\alpha$ = Crystal Structure 1<br>$\beta$ = Crystal Structure 2<br>$\alpha'$ = Superlattice within $\alpha$<br>Subscripts indicate different compositions |                             |                                  |



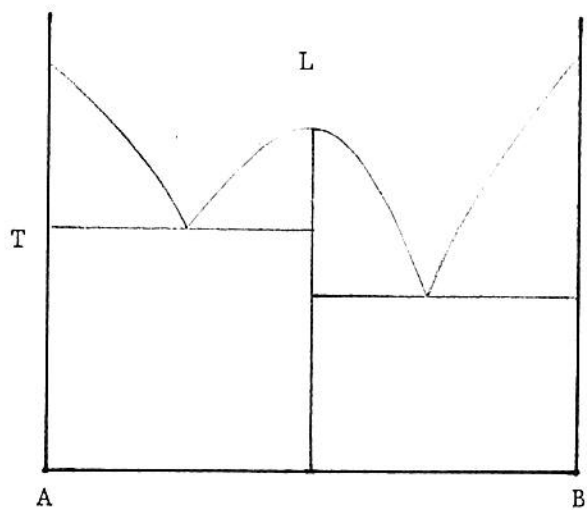
Simple Eutectic



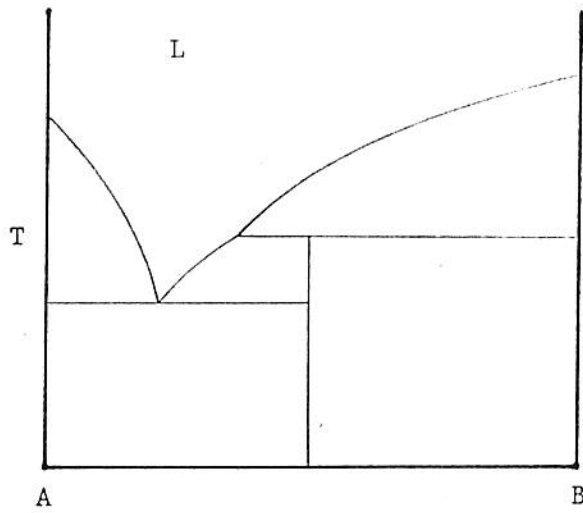
Subsolidus Polymorphism



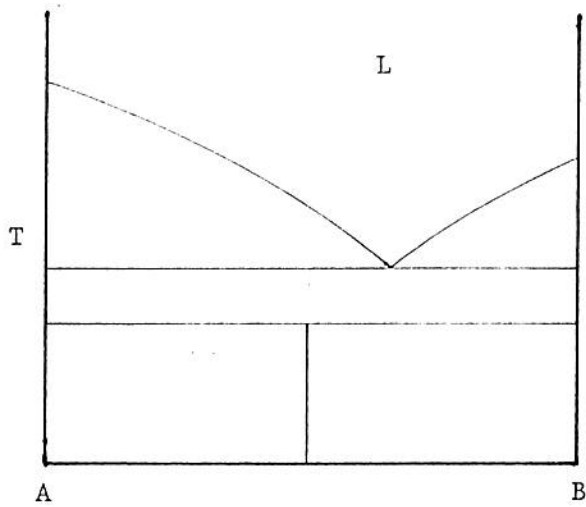
Supersolidus Polymorphism



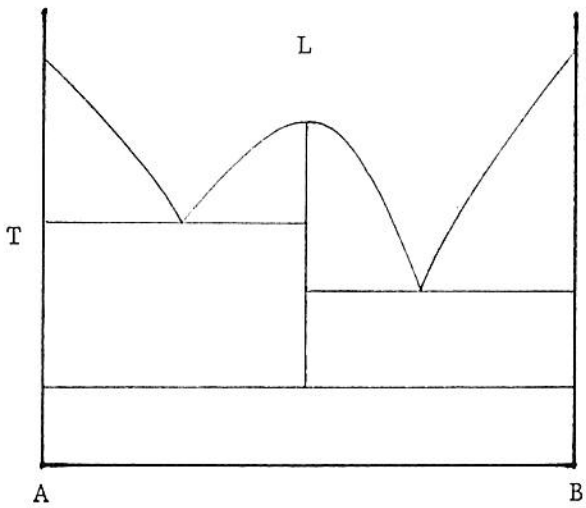
Congruently Melting Intermediate Compound



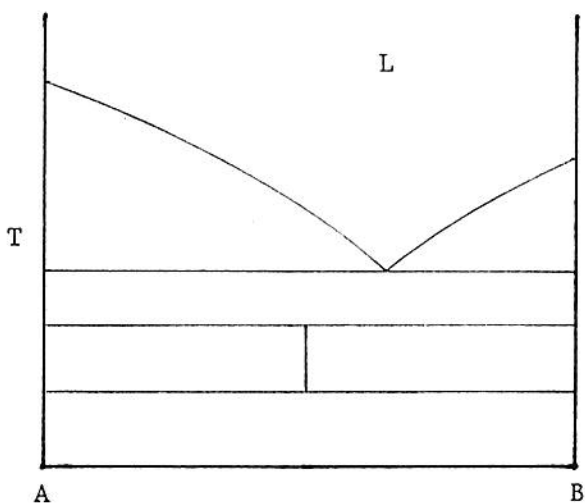
Incongruently Melting Intermediate Compound  
(Peritectic, Meritectic)



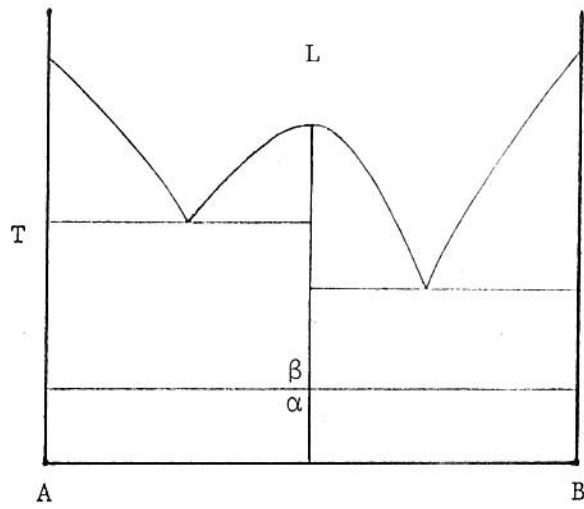
Dissociation with Upper Temperature Limit of  
Stability



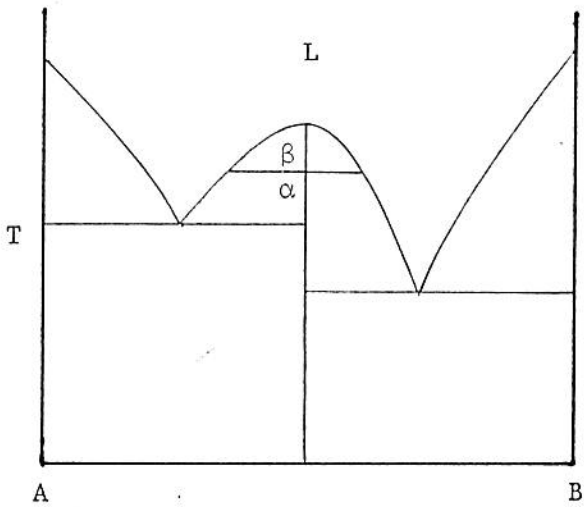
Dissociation with Lower Temperature Limit of  
Stability



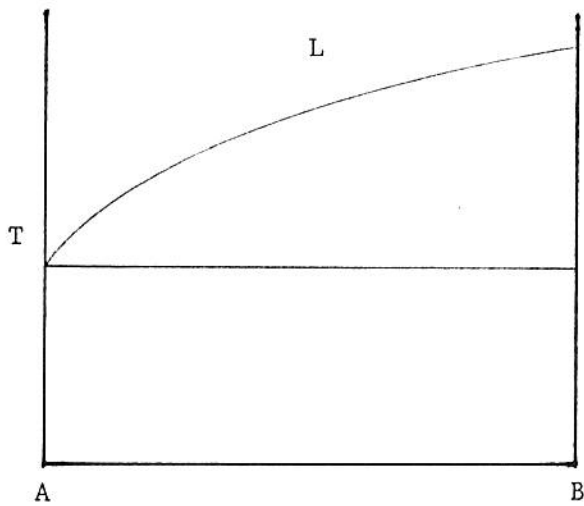
Dissociation with Upper and Lower Temperature  
Limits of Stability



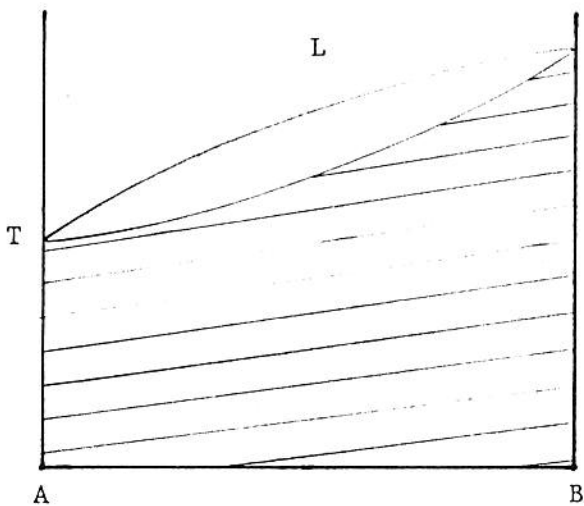
Subsolidus Polymorphism of Intermediate Compound



Supersolidus Polymorphism of Intermediate Compound

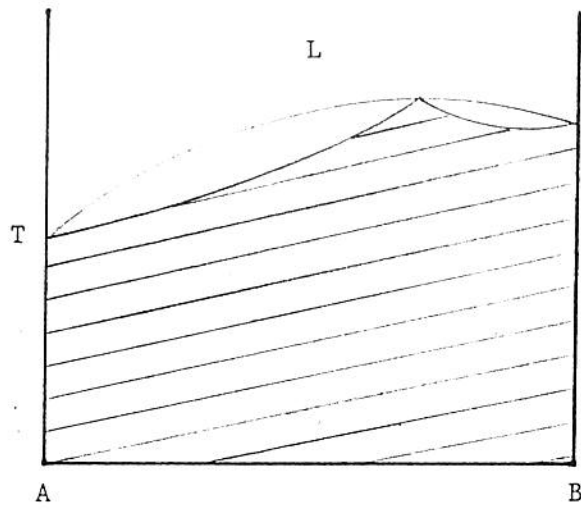


Monotectic (Disused Terminology)

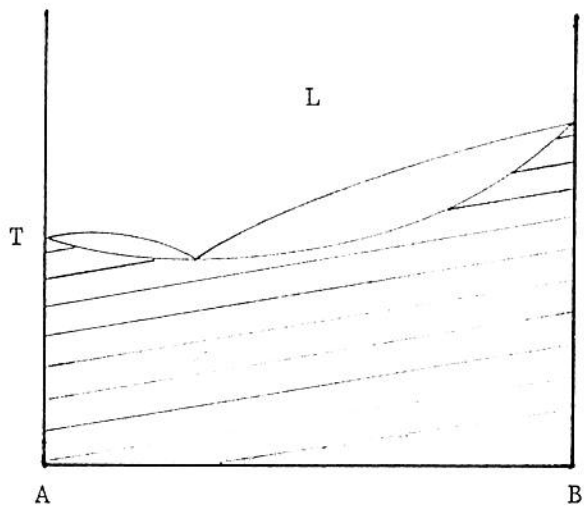


Continuous Primary Solid Solution

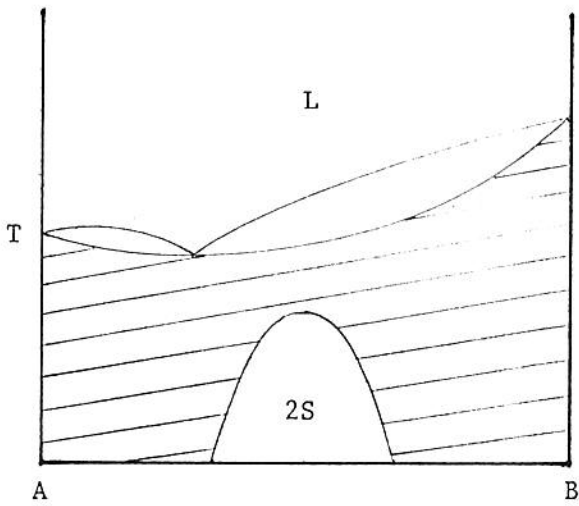




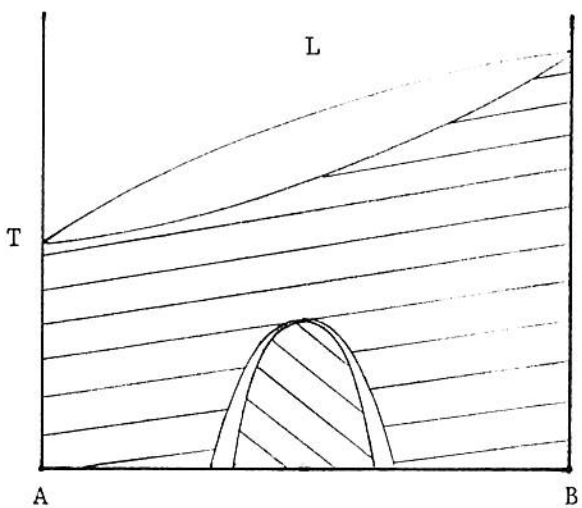
Maximum Melting Point Primary Solid Solution



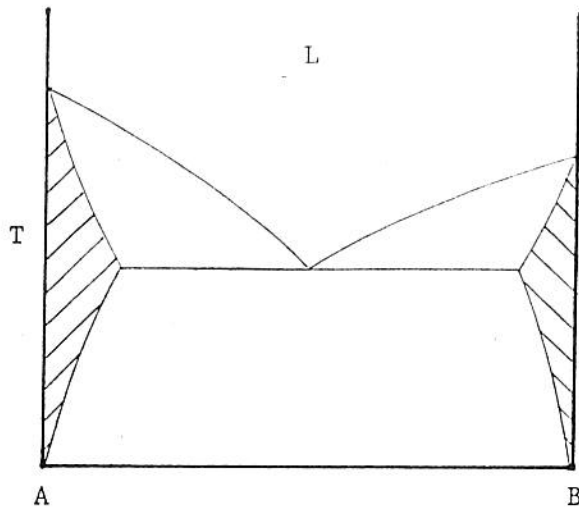
Minimum Melting Point Primary Solid Solution



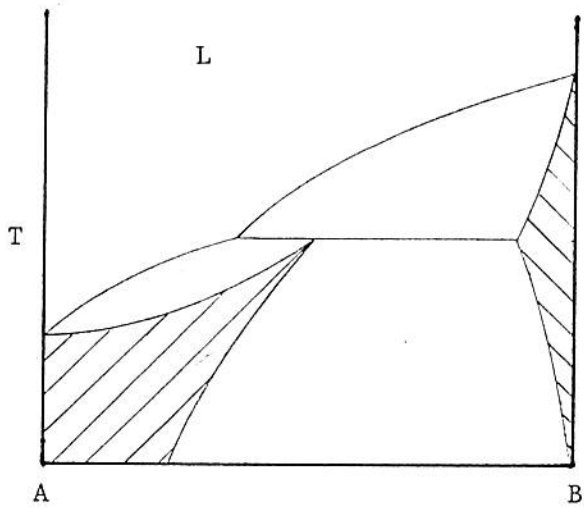
Completely Miscible Primary Solid Solution  
with Miscibility Gap (Exsolution)



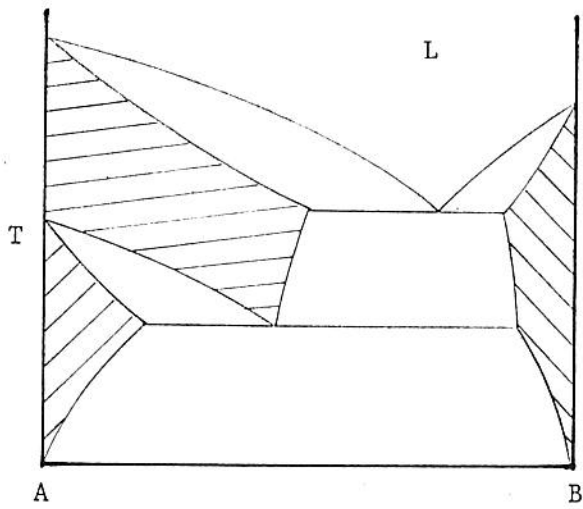
Completely Miscible Primary Solid Solution  
with Superlattice



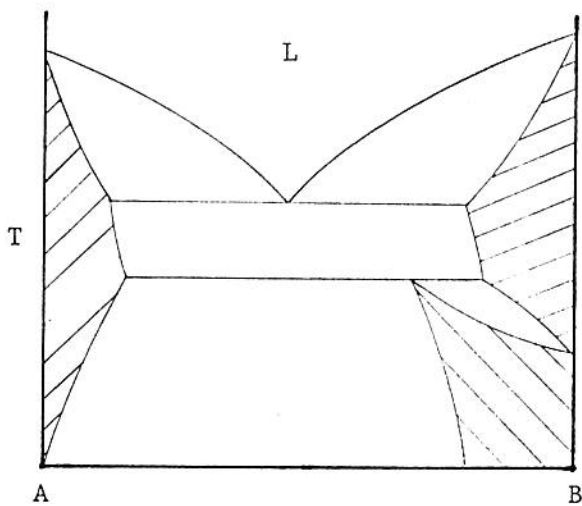
Partial Miscibility with Eutectic



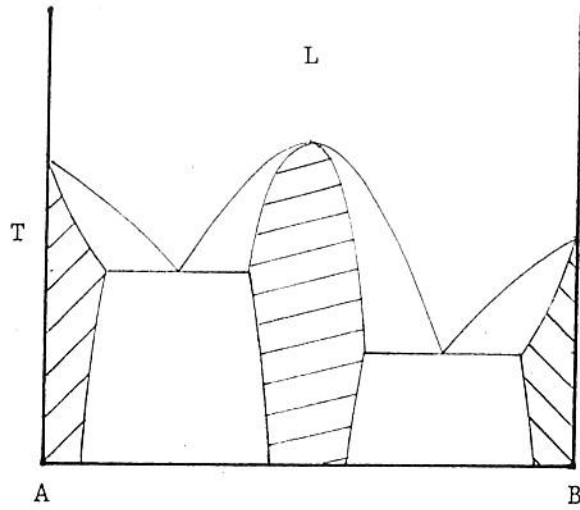
Partial Miscibility with Peritectic



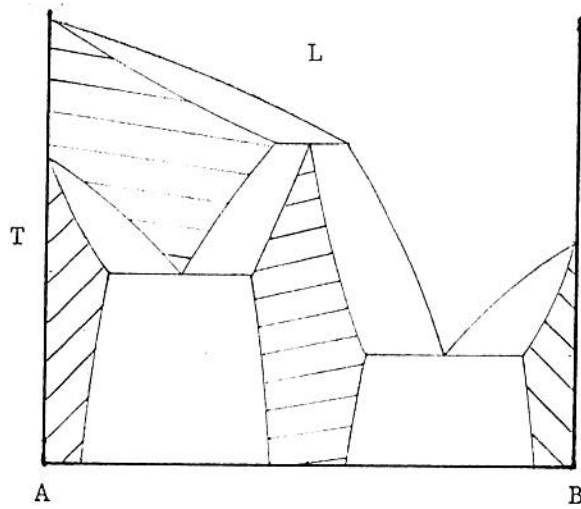
Partial Miscibility with Eutectoid



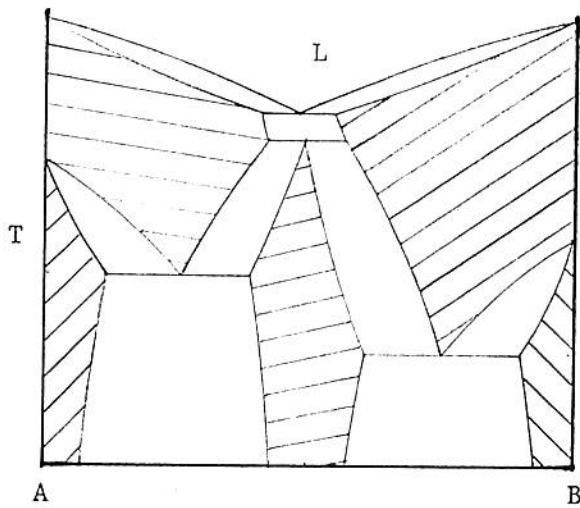
Partial Miscibility with Peritectoid



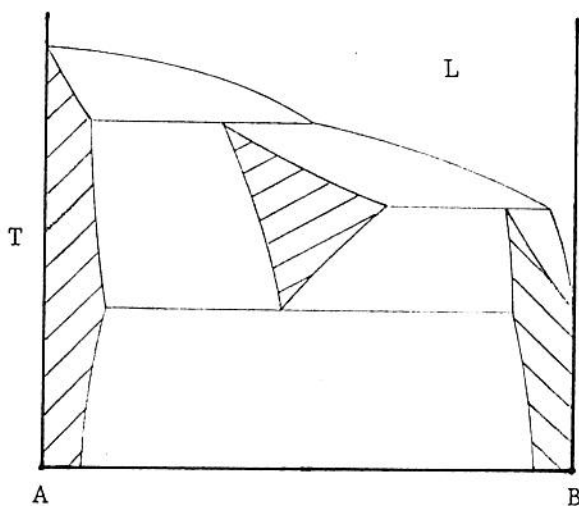
Partial Miscibility with Congruently Melting Intermediate Compound



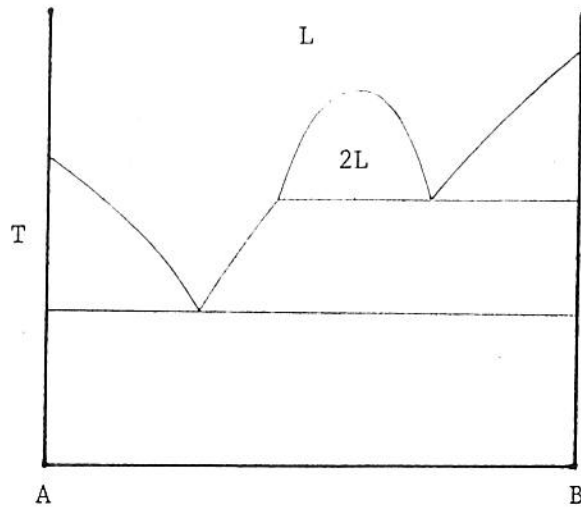
Partial Miscibility with Incongruently Melting Intermediate Compound



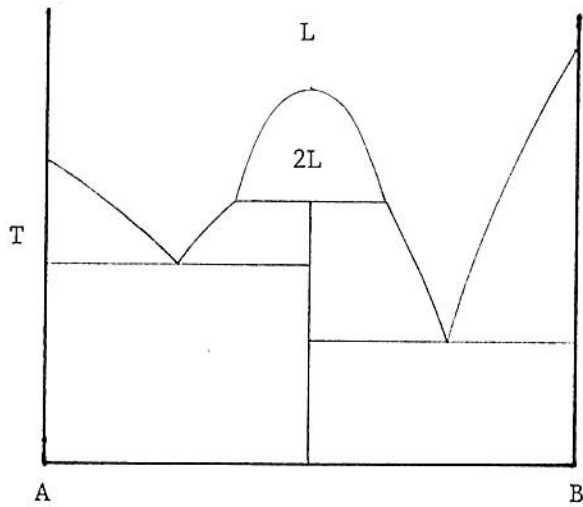
Partial Miscibility with Peritectoid Reaction of Intermediate Compound



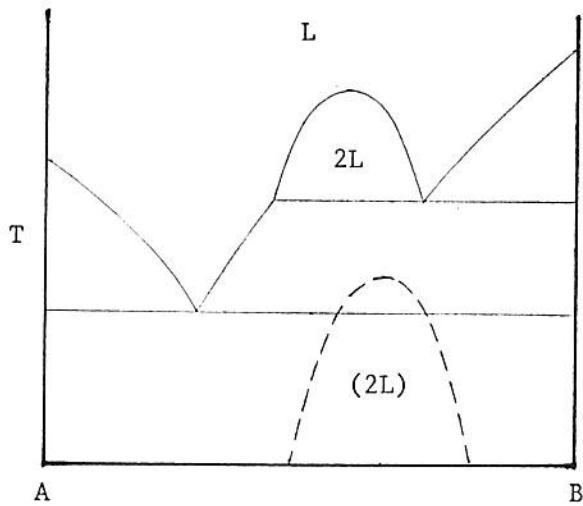
Secondary Solid Solution



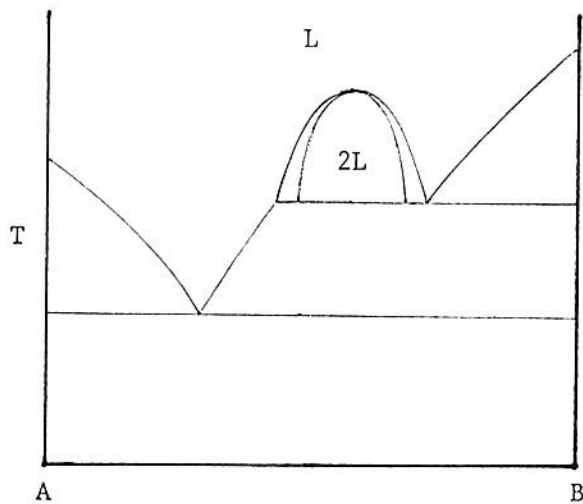
Liquid Immiscibility with Monotectic



Liquid Immiscibility with Syntectic



Metastable Liquid Immiscibility



Spinodal Decomposition

## Reference

R.A. Higgins, *Engineering Metallurgy. Part I: Applied Physical Metallurgy, Sixth Edition* (1993).

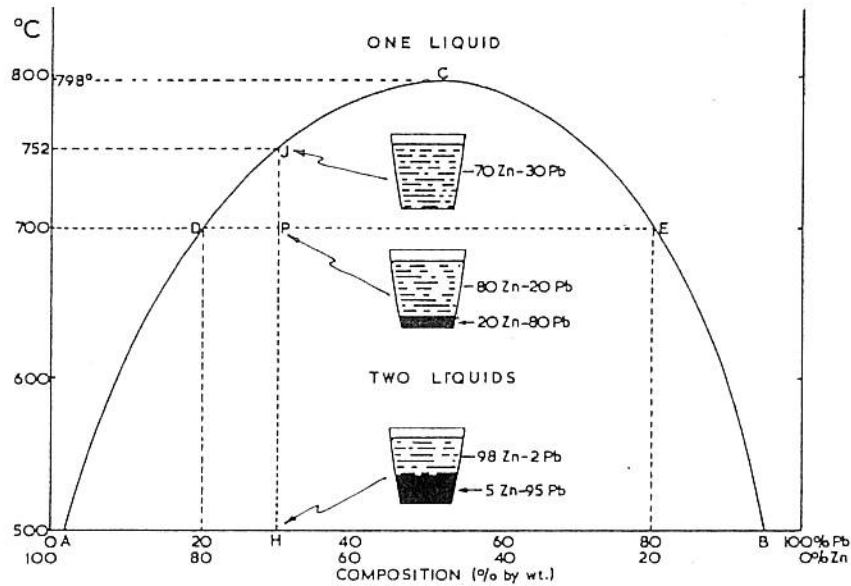


Fig. 9.7 The zinc-lead thermal equilibrium diagram above 500°C.

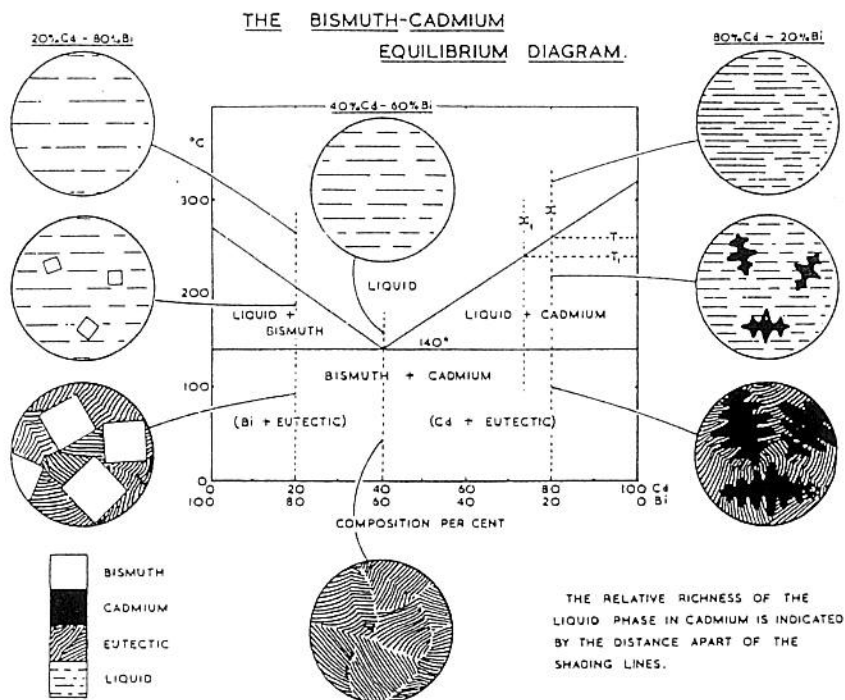
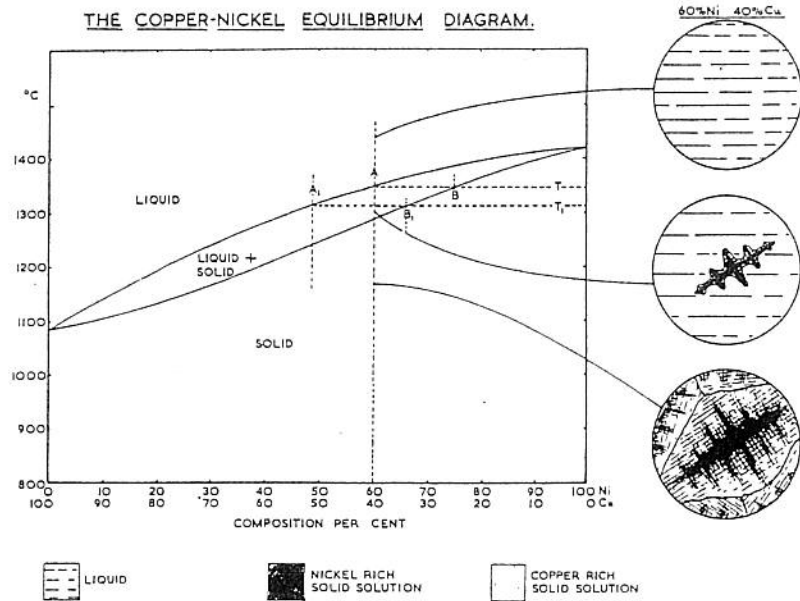
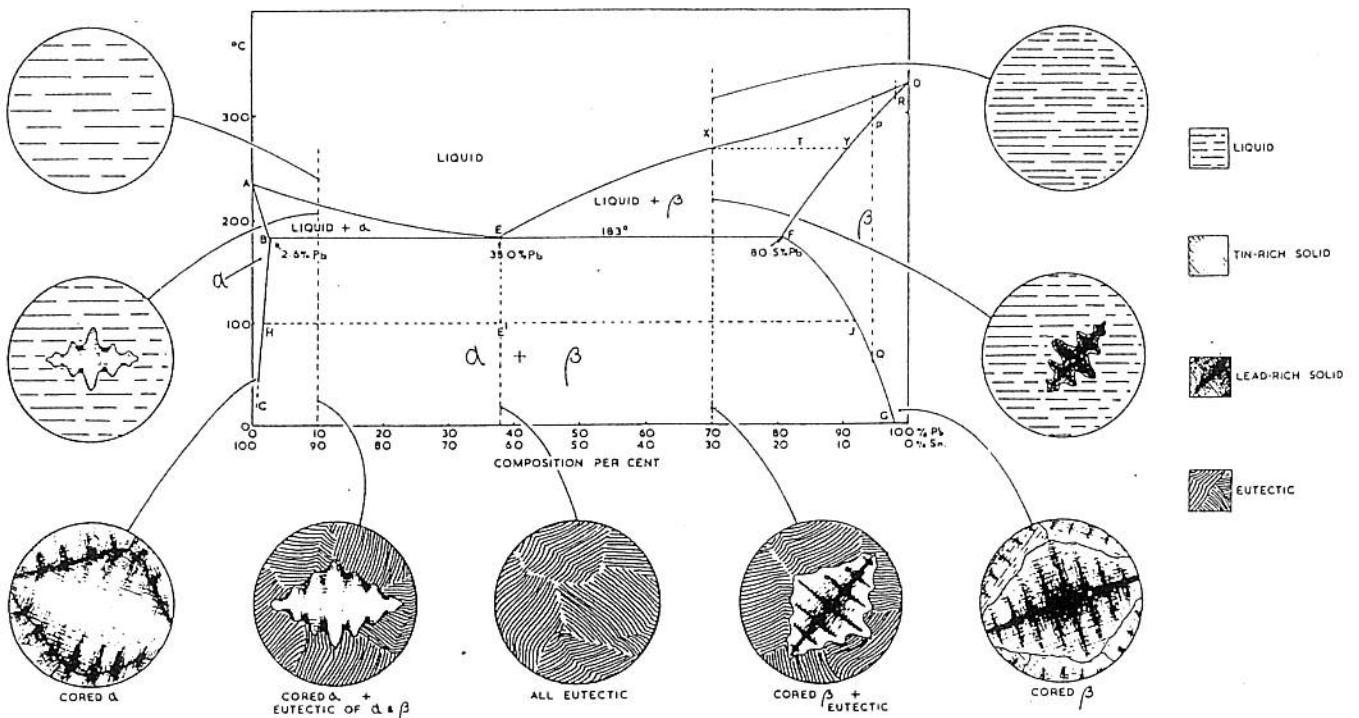


Fig. 9.8 The bismuth-cadmium equilibrium diagram.



**Fig. 9.9** The copper-nickel equilibrium diagram.  
The microstructures indicated are those obtained under *non-equilibrium* conditions of solidification.



**Fig. 9.12** The tin-lead equilibrium diagram  
The microstructures indicated are those obtained under *non-equilibrium* conditions of solidification.

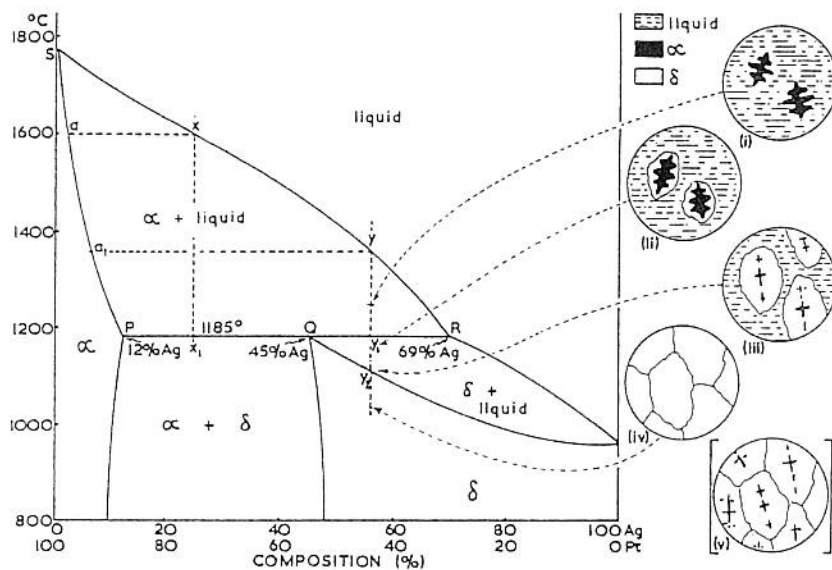


Fig. 9.14 The platinum-silver thermal equilibrium diagram.

(i) A slowly cooled alloy containing about 75% silver will first begin to solidify as uniform  $\alpha$ . (ii) Below 1185°C these  $\alpha$  crystals will become coated with  $\delta$  due to peritectic interaction with the remaining liquid. (iii) The original  $\alpha$  will be transformed slowly to  $\delta$  as the temperature falls. (iv) If cooling is *extremely slow* all  $\alpha$  will be absorbed and the remaining liquid will solidify as uniform  $\delta$ . (v) In practice cooling will generally be so rapid as to 'trap' some  $\alpha$  'cores' in the final structure, producing a non-equilibrium structure.

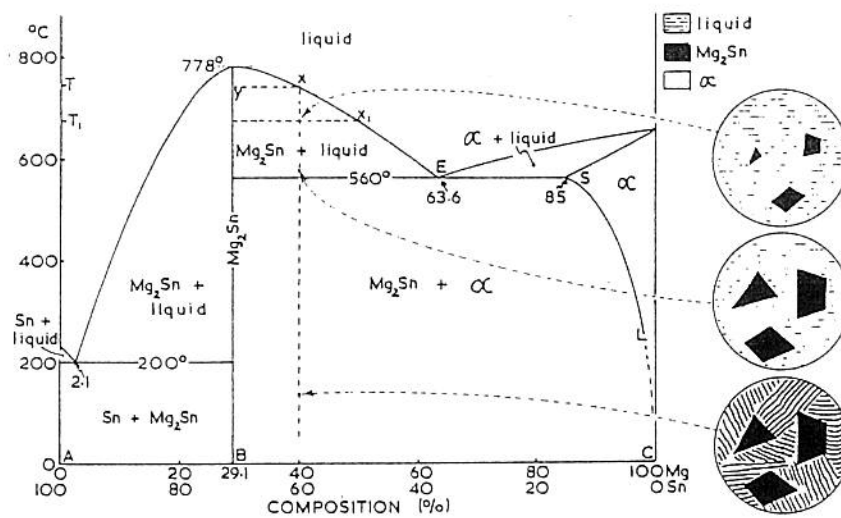


Fig. 9.15 The tin-magnesium thermal equilibrium diagram, consisting essentially of two eutectiferous series—tin- $\text{Mg}_2\text{Sn}$  and  $\text{Mg}_2\text{Sn}$ -magnesium.

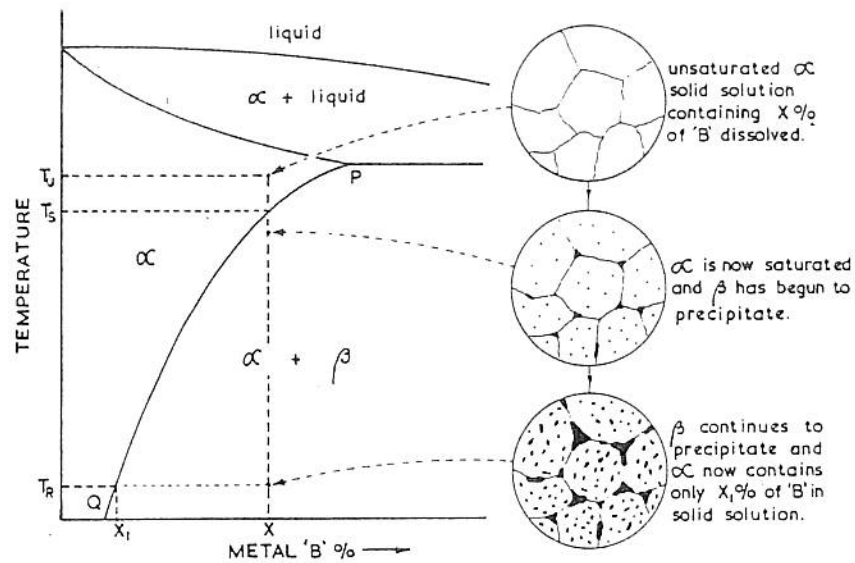


Fig. 9.18 Changes in partial solid solubility with temperature.



Plate 9.3 Representative tin-lead alloys in the cast condition.  $\times 100$ .

(i) 75 tin-25 lead. Primary crystals of  $\alpha$  (light) in a matrix of eutectic consisting of layers of  $\alpha$  (light) and  $\beta$  (dark).

(ii) 62 tin-38 lead. Completely eutectic in structure—layers of  $\alpha$  (light) and  $\beta$  (dark).

(iii) 50 tin-50 lead. Primary crystals of  $\beta$  (dark) in a matrix  $\alpha + \beta$  eutectic as in (i) and (ii). Note the small 'islands' of  $\alpha$  within the primary  $\beta$  crystals in (iii). These were precipitated as the alloy cooled slowly from  $183^\circ\text{C}$  to ambient temperature and the  $\beta$  changed in composition along the steeply sloping solvus  $FG$  (Fig. 9.12).



### THE CORING OF SOLID SOLUTIONS.

DIAGRAMS INDICATING THE RELATIVE DISTRIBUTIONS OF ATOMS OF METALS (A) & (B) AT VARIOUS STAGES DURING THE SOLIDIFICATION AND ANNEALING OF A SOLID SOLUTION CONTAINING 50% OF EACH METAL.

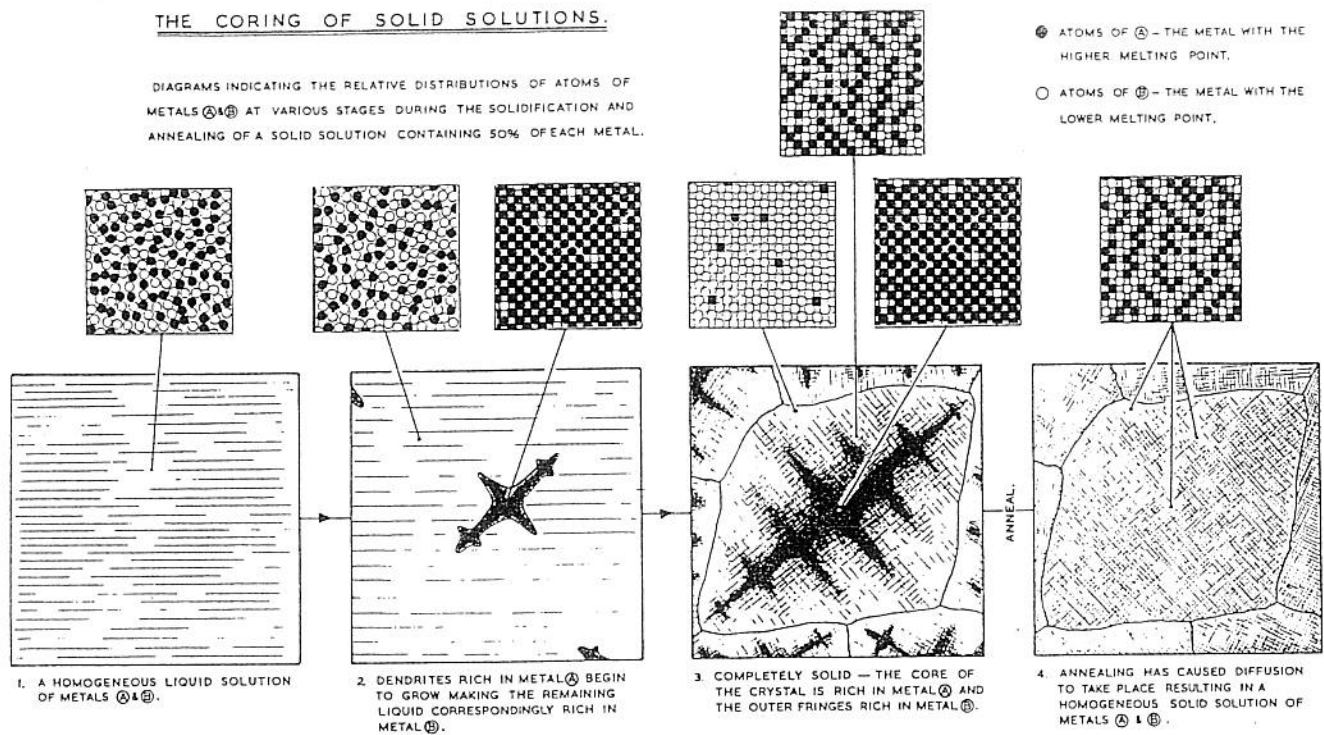


Fig. 8.2 The variations in composition which are possible in a solid solution.

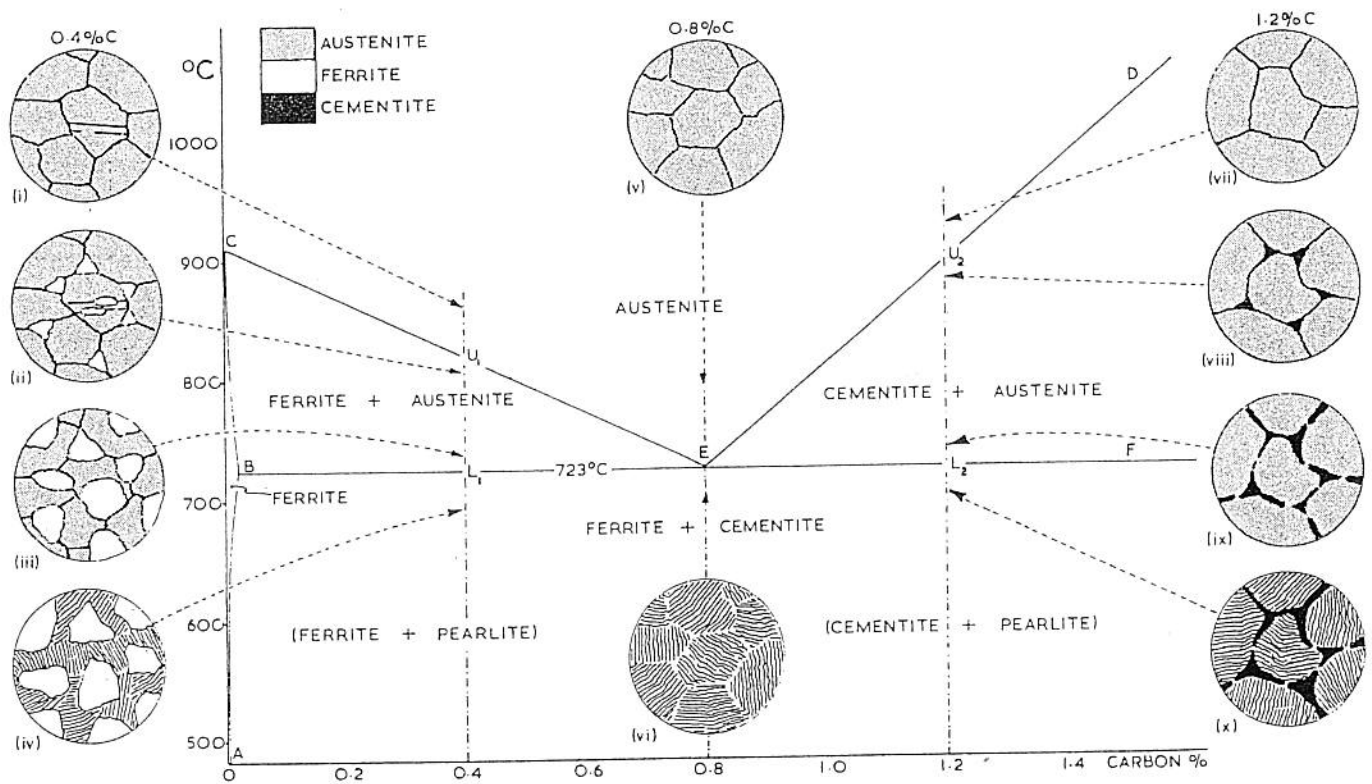


Fig. 7.6 Part of the iron-carbon thermal equilibrium diagram.

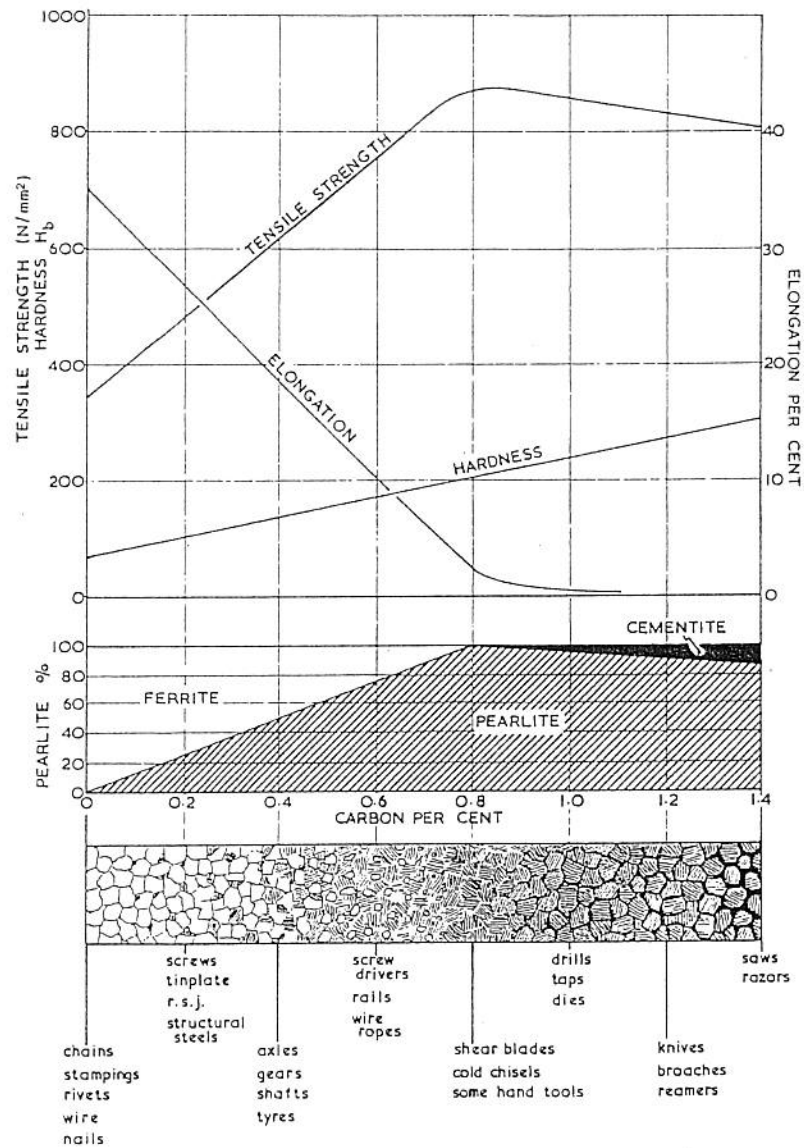


Fig. 7.7 A diagram showing the relationship between carbon content, microstructure and mechanical properties of plain carbon steels in the *normalised* condition. Typical uses of these steels are also indicated.